

Epic 3: Dataset Download and Understanding

Reading the Dataset:

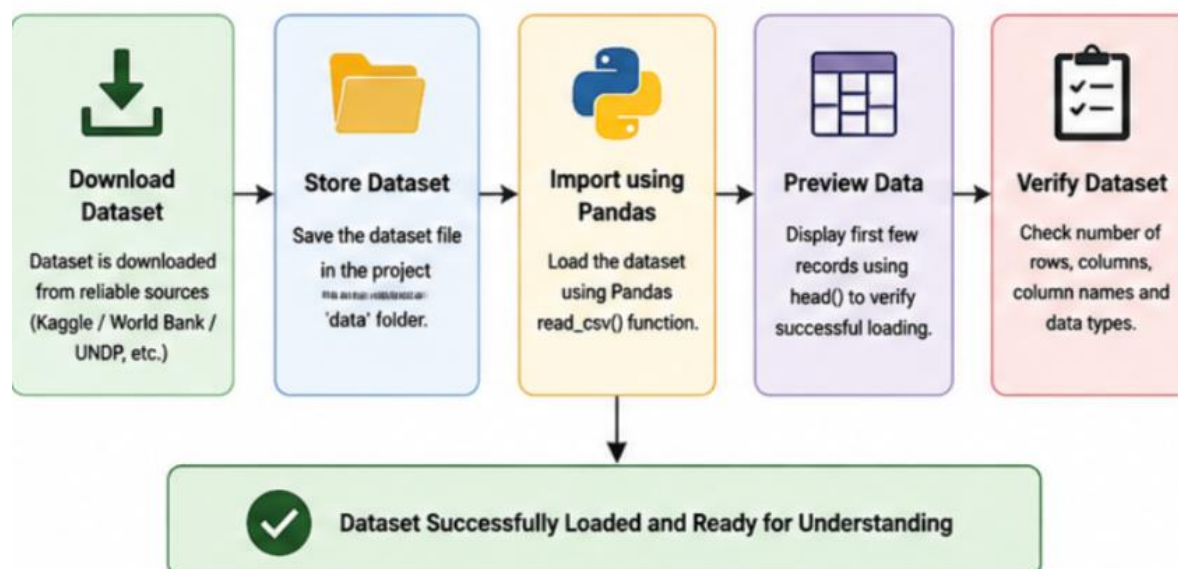
The first step in the Comprehensive Measure of Well Being project is downloading and reading the dataset. The dataset contains valuable information about various indicators such as health, education, income, employment, life expectancy, happiness, and economic conditions. These indicators help in evaluating the overall well-being of different countries or regions.

The dataset is collected from reliable sources like Kaggle, the World Bank, or the United Nations Development Programme (UNDP). After downloading, the dataset is stored in the project's data folder for organized management.

Using the Pandas library, the dataset is loaded into Python. Functions such as `read_csv()` are used to import the dataset into a Data Frame. The first few rows are displayed to verify successful loading. The number of rows, columns, column names, and data types are also checked to ensure the dataset is ready for analysis.

This process ensures that the dataset is complete, accurate, and properly imported before moving to the next phase.

1. DATASET READING WORKFLOW



Understanding the Dataset:

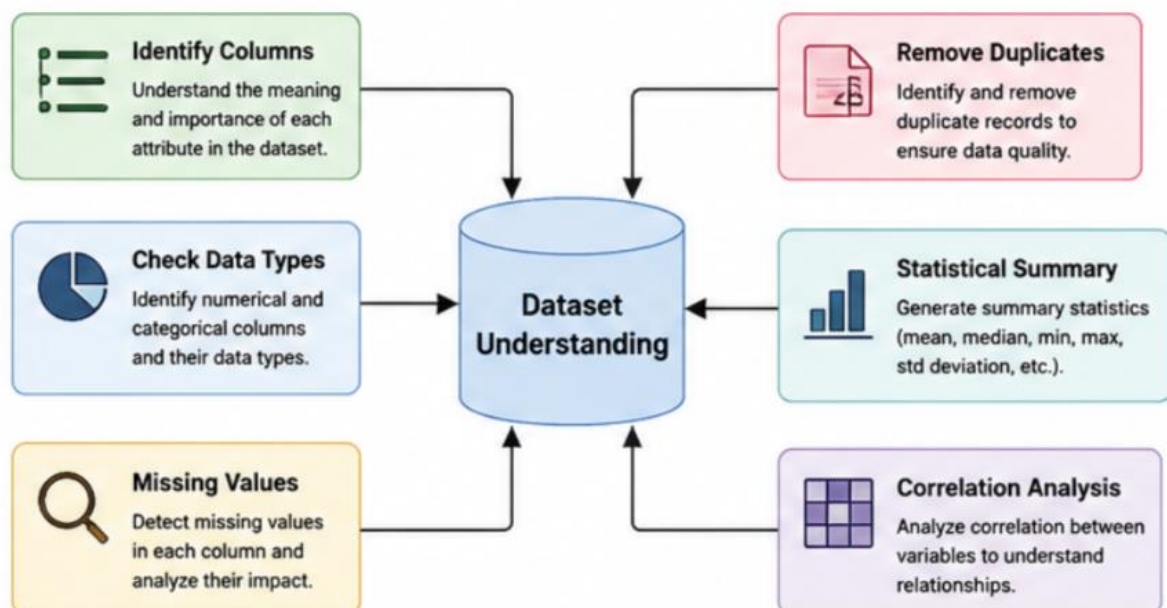
After loading the dataset, the next step is to understand its structure and contents. Each column represents a specific indicator related to human development and well-being. Numerical attributes include values such as GDP per capita, life expectancy, education index, and happiness score, while categorical attributes include country names and regions.

The dataset is carefully examined to identify missing values, duplicate records, and incorrect data types. Statistical summaries including mean, median, minimum, maximum, and standard deviation are generated to understand the distribution of data.

Understanding the dataset helps identify important variables that significantly contribute to measuring overall well-being. This stage also helps determine the cleaning and preprocessing steps required before building predictive models.

A clear understanding of the dataset improves the quality and reliability of the analysis performed in later stages.

2. DATASET UNDERSTANDING PROCESS



Data Visualization:

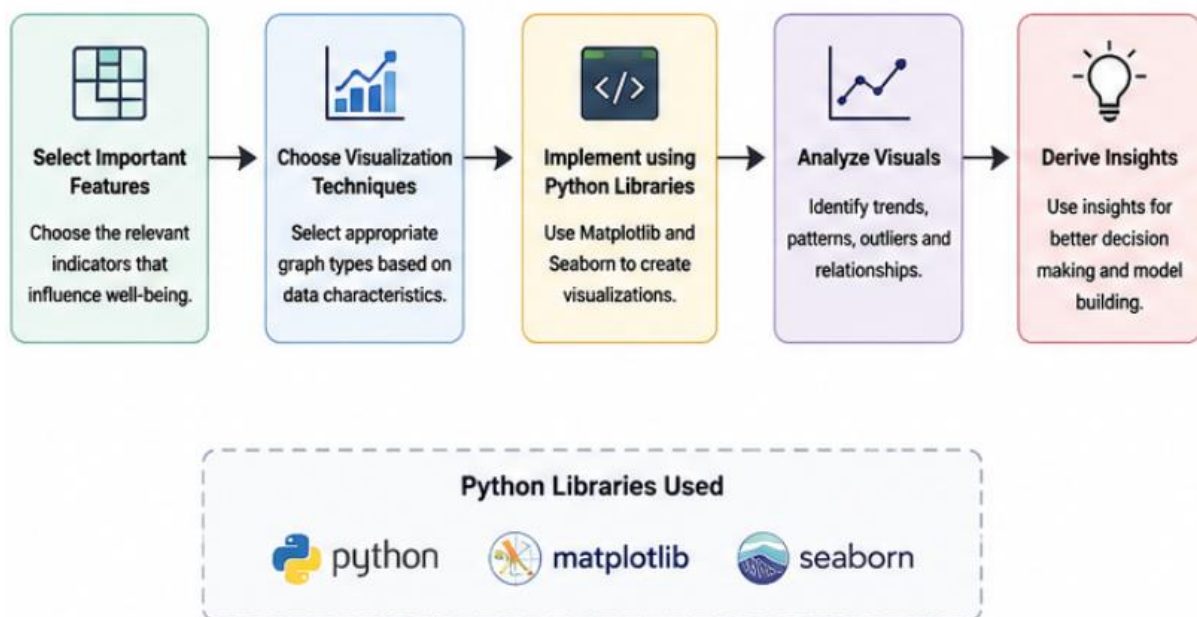
Data visualization is an essential part of data analysis because it converts numerical information into graphical representations that are easier to understand. Visualization helps identify trends, patterns, relationships, and outliers within the dataset.

Python libraries such as Matplotlib and Seaborn are used to generate various charts and graphs. Common visualization techniques include bar charts, histograms, scatter plots, box plots, pie charts, and correlation heatmaps.

These visualizations help compare countries based on different well-being indicators, analyze the distribution of variables, and understand correlations between multiple features.

The insights obtained through visualization support better decision-making during feature selection and preprocessing, making the machine learning model more effective.

3. DATA VISUALIZATION WORKFLOW



Outcome and Conclusion:

After completing dataset reading, understanding, and visualization, the project team gains a comprehensive understanding of the available information. The dataset structure becomes clear, important features are identified, and data quality issues such as missing values and duplicates are recognized.

Visual analysis provides meaningful insights into relationships among different well-being indicators and highlights trends across countries or regions.

The dataset is now organized, verified, and ready for preprocessing, feature engineering, and machine learning model development. This stage forms the foundation for building an accurate and reliable Comprehensive Measure of Well Being prediction system.

4. TYPES OF VISUALIZATIONS USED

